School of Computer Science – Coursework Issue Sheet

Session	2026-2027	**Semester**	Autumn
Module Name	Natural Language Processing	**Code**	COMP4153
Module Convenor(s) **(CW Convenor in Bold)**	Chin Poo Lee, Kian Ming Lim		

Coursework Name	Coursework 1	**Weight**	70%

Deliverable (a brief description of what is to be handed-in; e.g. "software", "report", "presentation", etc.)	This is an individual NLP research project. You will define an NLP research problem, compare at least five distinct recent methods on at least two datasets, critically analyse the results, and propose, implement and evaluate your own approach/model with a clearly identifiable research contribution. The final written output should take the form of an academic research paper. **Submit:** • **Research paper:** one IEEE-style two-column PDF covering all required tasks. • **Source code:** at least twelve executed Jupyter Notebook files (`.ipynb`) submitted in one ZIP file, corresponding to the required method-dataset experiments.
Format (summary of the technical format of deliverable, e.g. "C source code as zip file", "pdf file, 2000 word max", "ppt file, 10 slides max", etc.)	Submit the research paper and source code separately via Moodle. **1. Research Paper** Submit `Name_ID.pdf` (e.g. `XiaomingChen_20712345.pdf`). Use the official IEEE Conference template in conference mode, standard two-column layout, A4 paper size, and default IEEE font sizes, margins and spacing. Either of the official template formats may be used to prepare the paper. The main paper must not exceed **8 pages, excluding references and the mandatory appendix**. Do not alter the template formatting to fit additional content. After the references, include one appendix page for each required method-dataset combination. This gives a minimum of **12 appendix pages**: • at least five existing methods $\times$ two datasets ≥ 10 pages; and • one proposed method $\times$ two datasets $= 2$ pages. Each appendix page must clearly identify the method and dataset and show the final experimental output produced by the corresponding notebook, including quantitative results and, where appropriate, representative qualitative outputs. The appendix is for experimental evidence only and does not count towards the 8-page limit. **2. Source Code** Submit `Name_ID_source.zip` (e.g. `XiaomingChen_20712345_source.zip`) containing at least **12 Jupyter Notebook files**: • at least 10 notebooks for the existing methods on the two datasets; and • 2 notebooks for the proposed method on the two datasets. If more than five existing methods are evaluated, include the corresponding additional notebooks. Name each notebook clearly, for example: `Name_ID_MethodName_DatasetName.ipynb`. Each notebook must: • reproduce one method on one dataset and run independently from top to bottom; • include setup cells that check required Python packages and install missing packages where necessary, with important package versions stated; • include the relevant data processing, random seed(s), model setup, training/evaluation and result generation; and • be submitted with executed outputs retained, including the main quantitative results and any relevant tables/figures. Excessively verbose logs may be suppressed. Do not include datasets, pretrained model weights, cache files, separate `.py` files, `README` files, `requirements.txt`, `environment.yml`, or other source-code files in the submission.
Acceptable Use of Artificial Intelligence (AI) Examples of AI tools include large language models (e.g. ChatGPT, Claude, Kimi), AI coding assistants (e.g. GitHub Copilot) and image, audio or video generators (e.g. DALL-E, Midjourney, Suno). This is not an exhaustive list.	🟡 **Amber:** AI may be used for the following activities when completing this coursework: supporting learning and understanding of relevant concepts, idea organisation, grammar checking, and limited programming assistance such as debugging, syntax support, and explanation of library/API usage. AI-generated content must not replace the student's own research thinking, analysis, experimental interpretation, or research contribution. All other use of AI is strictly prohibited. The submitted research paper must include a clear AI usage declaration stating either that no AI tools were used, or identifying the AI tool(s) used and briefly explaining how they were used in preparing the coursework. Students remain fully responsible for the accuracy, originality and integrity of all submitted work.

Issue Date	12 October 2026
Submission Date	18 December 2026, 15:00
Submission Mechanism	Online submission via Moodle. You do NOT need to submit a hardcopy of your report to the faculty office.
Late Policy (University of Nottingham default will apply, if blank)	The standard UNNC late submission policy applies.
Feedback Date	29 January 2027
Feedback Mechanism	Online feedback on Moodle.

Instructions

Conduct an individual, small-scale research study addressing a clearly defined Natural Language Processing (NLP) problem. The study should involve implementing and comparing recent methods, developing a proposed approach, conducting rigorous and reproducible experiments, and critically analysing the findings. Topics primarily focused on non-NLP tasks, such as image recognition, image-text matching without substantial language analysis, knowledge graph link prediction using only structured data, or tabular data prediction, are not suitable. Interdisciplinary approaches are acceptable provided that NLP remains the primary research focus. The work should be presented as an academic research paper with a clear research question, methodology, contribution, experimental results and evidence-based conclusions.

Tasks and Required Research Paper Content

Complete all six tasks below. The tasks follow the normal structure of an academic research paper and should be integrated into one coherent manuscript rather than written as six independent answers.

Task 1. Title, Abstract and Introduction.

- Provide a clear and informative research title.
- Write a concise abstract summarising the NLP problem, datasets, comparison methods, proposed approach/model, main experimental findings and principal conclusion.
- Introduce the research problem, motivation and scope (including the task input and expected output), and formulate a clear research question or objective.
- Explain why the problem is important and state the main contribution(s) of the paper clearly.

Task 2. Related Work and Research Gap.

- Review relevant and sufficiently recent research to establish the current state of work on the selected NLP problem. The coverage should be appropriate to the topic and broad enough to position the study within the existing literature.
- Organise the review around the main ideas, methodological families, representative approaches and findings that are most relevant to the research problem, rather than presenting a sequence of disconnected paper summaries.
- Critically compare relevant approaches in terms of their assumptions, methodology, strengths, limitations and reported evidence. Explain how the literature relates to the research question and to the experimental methods selected later in the paper.
- Use authoritative academic sources and cite the original papers for the principal methods or ideas discussed wherever possible.
- Identify a clear research gap, limitation, unresolved issue or opportunity emerging from the literature, and use it to motivate and position the student's proposed approach/model and own research contribution.

Task 3. Proposed Method and Own Contribution.

- Develop and present a **proposed approach/model** for the selected NLP task. The proposed approach/model is required and should embody a clearly identifiable contribution made by the student.
- Clearly explain what the student's own research contribution is and distinguish it from the ideas, components or resources adopted from existing work.
- The proposed approach/model should be clearly motivated by the research problem, related literature, identified research gap and/or limitations observed in existing approaches.
- The contribution must be substantive rather than merely changing model size, checkpoint, prompt wording, learning rate or another routine hyperparameter.
- Describe the proposed approach/model clearly and explain its main components and differences from relevant existing approaches. Use an appropriate **architecture/workflow figure, algorithm and/or mathematical formulation** where applicable.
- Explain the rationale for the proposed design and why it is expected to address the identified research gap, limitation or research objective.

Task 4. Datasets, Experimental Setup and Baseline Implementation.

- Use at least **two public datasets for the same NLP task**, sufficiently different to support meaningful cross-dataset comparison.
- Include a concise **dataset summary table** covering dataset size, label/output space, data characteristics and relevant class/category distributions.
- Describe preprocessing/tokenisation, train/development/test splits, evaluation metrics, hardware/software environment and main training/evaluation settings.
- Implement or appropriately adapt established open-source implementations of at least **five existing methods** (published in recent 5 years) and evaluate **each method on both datasets**. Clearly cite external code, pretrained resources, libraries and model checkpoints, and explain important implementation changes made for comparability.
- Using lightweight and pretrained models are acceptable, provided that the work meets the coursework requirements and is supported by rigorous evaluation.
- Apply a fair and consistent protocol across methods. Report random seeds and repeated runs where feasible; if an expensive method is run once, state this limitation.
- Published results may be cited for context but do not replace the student's own experimental results.

Task 5. Results, Quantitative and Qualitative Analysis, and Discussion.

- Evaluate the proposed method on **both datasets** and compare it with the required existing methods using the same experimental protocol.
- Include at least one **main comparison table** showing at least five existing methods, the proposed method, both datasets and the principal evaluation metric(s).
- Use **quantitative evaluation** to support the main performance claims and include **qualitative evaluation** where it provides useful insight into model behaviour, outputs, strengths or failure cases.
- Analyse important differences, consistencies and trade-offs across methods and datasets, relating them to relevant

methodological or data characteristics rather than merely ranking methods by one metric.
- Conduct an **error analysis of the proposed method** using representative successes and failures, identify plausible causes and discuss relevant limitations.
- Conduct an **ablation or component analysis** where applicable, or another suitable controlled investigation, to examine the contribution of the proposed methodological components.
- Critically interpret the evidence and relate the findings back to the research question, related work and claimed contribution. A performance improvement alone is not sufficient for high marks.

Task 6. Conclusion, References and Reproducibility.
- Summarise the main findings and explain how they answer the research question or objective.
- Clearly state the strengths, limitations and possible future directions of the proposed method and the overall study.
- Ensure all papers, datasets, external code, pretrained models and other resources are appropriately cited and referenced.
- Document the implementation choices, experimental settings and computational constraints needed to reproduce the main results.
- Ensure that results reported in the paper are traceable to and consistent with the corresponding submitted notebook outputs and appendix evidence.
- Present the complete work as a coherent academic research paper with clear organisation, professional figures/tables and evidence-based conclusions.

Recommended Research Paper Structure

The six tasks above are intended to map directly onto a conventional academic research paper. The exact section titles may be adapted to the topic, but the following content should normally be included:

1. **Title and Abstract** — provide a clear research title and a concise summary of the research problem, objective, datasets, methods/approach, key experimental results and main conclusion.
2. **Introduction** — introduce and motivate the NLP problem, define the scope, state the research question or objective, and clearly summarise the main contribution(s) of the work.
3. **Related Work** — critically synthesise relevant and sufficiently recent literature, explain the main approaches and their strengths/limitations, establish the context for the study, and identify the research gap or unresolved issue that motivates the work.
4. **Proposed Method** — describe the proposed approach/model and the student's own research contribution, explain its motivation and main components, distinguish it from relevant existing approaches, and support the description with an appropriate architecture/workflow figure, algorithm and/or mathematical formulation where applicable.
5. **Datasets and Experimental Setup** — describe the datasets, preprocessing, data splits, evaluation metrics, comparison methods, implementation details, training/evaluation settings, computational environment and information required for fair and reproducible experiments.
6. **Results and Discussion** — present the main quantitative comparison across methods and datasets, include qualitative analysis where it helps interpretation, critically explain performance differences and trade-offs, and include appropriate error analysis and ablation/component or other controlled analysis of the proposed approach/model.
7. **Conclusion** — summarise the principal findings, answer the research question or objective, discuss the main limitations, and identify appropriate directions for future work.
8. **References** — cite all academic papers, datasets, external code, pretrained models, software resources and other sources used in the work using the required IEEE referencing style.
9. **Appendix** — provide one page of experimental output for each required method-dataset combination (minimum 12 pages for five existing comparison methods plus the proposed approach/model across two datasets). The appendix is experimental evidence and does not count towards the 8-page main-paper limit.

Additional Requirements
- The work must be your own and comply with the UNNC Academic Integrity Policy. Clearly acknowledge and cite all external code, pretrained models, datasets and other resources.
- Plagiarism, collusion and submitting the same or substantially similar work for multiple assessments are prohibited.
- Clearly distinguish your own experimental results from values quoted from published work.
- Figures and tables must be legible in the IEEE two-column layout, numbered, captioned and discussed in the text.
- Discussion, interpretation, qualitative analysis, error analysis and ablation/controlled analysis required for assessment must remain in the main paper and must not be moved to the appendix to bypass the 8-page limit.
- The paper should prioritise experimental reasoning, critical analysis and the student's own contribution. Merely running five existing methods and presenting a performance table is insufficient for high marks.

Assessment Criteria

The coursework is marked out of **100 marks** and contributes **70%** to the overall module mark. All six tasks are required. The assessment follows the structure of the research paper. Marks reward research quality, technical correctness, critical analysis, the significance and validation of the student's own contribution, reproducibility and academic presentation rather than the mere presence of a required section, table or figure.

Task	Marks	Criteria
1. Title, Abstract and Introduction	10	• Clear research title, well-focused NLP problem, motivation, scope and research question/objective. • Abstract accurately summarises the problem, datasets, methods, proposed contribution, principal results and conclusion. • Main contribution(s) are stated clearly and the importance of the study is convincingly established.
2. Related Work and Research Gap	10	• A suitable range of relevant and sufficiently recent literature is selected and supported by authoritative academic references. • Literature is critically synthesised, with meaningful comparison of methodological ideas, strengths, limitations and reported evidence rather than a sequence of paper summaries. • A credible research gap, unresolved issue or limitation is identified from the literature and used to position and motivate the proposed work and the student's own contribution.
3. Proposed Method and Own Contribution	30	• The proposed approach/model is clearly motivated, technically sound and distinguishable from existing work. • The student's own contribution is substantive rather than a routine model, prompt or hyperparameter change. • Method components are explained clearly using suitable architecture/workflow figures, algorithms and/or mathematical formulation where appropriate. • The design is logically connected to the identified research gap or limitation and demonstrates independent research thinking.
4. Datasets, Experimental Setup and Baseline Implementation	20	• Two suitable public datasets are appropriately selected and described. • Preprocessing, splits, metrics, implementation details and experimental settings are technically sound and sufficiently clear. • At least five existing methods are correctly implemented/adapted and evaluated on both datasets using a fair and consistent protocol. • External resources are properly cited and implementation choices, random seeds, repeated runs and relevant computational constraints are documented where appropriate.
5. Results, Quantitative/Qualitative Analysis and Discussion	20	• Results compare at least five existing methods and the proposed method across both datasets using appropriate quantitative metrics. • Qualitative examples or outputs are included where they add meaningful insight into model behaviour or performance. • Analysis explains performance differences, cross-dataset behaviour and relevant methodological/data trade-offs rather than merely restating results. • Error analysis and ablation/component analysis, or another suitable controlled investigation, provide evidence about the proposed contribution and its limitations. • Claims and conclusions are supported by the experimental evidence and related back to the research question and literature.
6. Conclusion, References and Reproducibility	10	• Conclusion clearly summarises the findings, answers the research question and discusses limitations and future work. • Academic references and external resources are cited accurately and consistently. • Submitted notebooks and appendix evidence reproduce and support the main results reported in the paper. • The final paper is coherent, professionally organised and presented to an appropriate academic research-paper standard.
Total	**100**	